Toward Transparent Virtual Coupling for Haptic Interaction during Contact Tasks *

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ABSTRACT

Since its introduction (e.g., [4, 6]), virtual coupling technique has been *de facto* way to connect a haptic device with a virtual proxy for haptic rendering and control. However, because of the single dependence on spring-damper feedback action, this virtual coupling suffers from the degraded transparency particularly during contact tasks when large device/proxy-forces are involved. In this paper, we propose a novel virtual coupling technique, which, by utilizing passive decomposition, reduces device-proxy position deviation even during the contact tasks while also scaling down (or up) the apparent inertia of the coordinated device-proxy. By doing so, we can significantly improve transparency between multiple degree-of-freedom (possibly nonlinear) haptic device and virtual proxy. In other to use passive decomposition, disturbance observer of [3] is adopted to estimate human force with some dead-zone modification to avoid "winding-up" force estimation in the presence of device-torque saturation. Some preliminary experimental results are also given to illustrate efficacy of the proposed technique.

Index Terms: virtual coupling, passive decomposition, transparency, contact task, multi-link virtual proxy, passivity, disturbance observer

1 INTRODUCTION

Due to its simplicity and intuitiveness, virtual coupling [1, 4, 6] has been the *de facto* way in haptics to connect an haptic device and a virtual proxy in the virtual world. See also [5, 11, 12, 13, 15, 22] for some applications and theoretical results related to this virtual coupling technique. Even with this wide usage and dominant presence in haptics, however, virtual coupling technique suffers from the degraded transparency during contact tasks because it solely depends on the spring-damper feedback action with the lack of coordinated feedforward cancelation of the human and virtual proxy forces. Here, recall that the ideal transparency [24] requires *both* the perfect position coordination between the haptic device and the virtual proxy and also the zero inertia of the coordinated haptic device and virtual proxy. In fact, virtual coupling needs the very position coordination error between the haptic device and the virtual proxy to produce haptic feedback during contact tasks.

In this paper, we aim to improve this transparency of the virtual coupling technique for haptic interaction. In particular, we consider the cases where the virtual proxy has (possibly nonlinear) Lagrangian dynamics with multiple degree-of-freedoms (DOFs) [17], which is able to provide much richer haptic experience than the usual point-mass type virtual proxies [1, 5, 6, 15]. The multi-DOF Lagrangian virtual proxy can allow for contacts with virtual objects/environments not only at its end-effector tip but also on *any parts* of its links. See Fig. 1. For the haptic device and this multi-DOF virtual proxy, both generally nonlinear, we then utilize and extend the passive decomposition approach [18, 19] to design a feedforward action, which can reduce the device-proxy position coordination error while also scaling down (or up) the apparent inertia of the coordinated haptic device and virtual proxy, thereby, significantly improve the transparency of the virtual coupling in haptic interaction particularly during contact tasks.

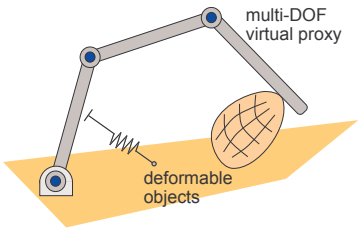

Figure 1: Multi-DOF Lagrangian virtual proxy [17].

More precisely, extending the passive decomposition [18, 19] to the hybrid-time haptics interfacing problem (i.e., continuous-time haptic device and discrete-time virtual proxy), we first split their (nonlinear) dynamics into the shape system (i.e., describing the device-proxy position-coordination aspect) and the locked system (i.e., abstracting their coordinated dynamics). We then design hybrid-time control to regulate the shape system to the origin with the cancelation of some (non-coordinating) portions of the human/proxy forces while also scaling down (or possibly up, resp.) the apparent inertia of the coordinated device and proxy by scaling up (or possible down, resp.) the combined human/proxy forces on the locked system.

However, the hybrid-time control design process, as stated above, requires us to sense (real) human force and (virtual) proxy force. Although the (simulated) force of the virtual proxy is always measurable, the human force sensing is typically unavailable in most commercially-available haptic devices (e.g., Force Dimension Omega3®, Geomagic Phantom Omni®, Novint Falcon®). To address this challenge, similar to [9], we utilize the disturbance observer presented in [3] to (sensorlessly) estimate the human force. We also propose a technique of using "dead zone" in [23] to mitigate the issue of "winding-up" in the estimated force accumulation in the presence of device control torque saturation. We also adopt the impulsive force generation technique similar to [7] to improve the perception of the contact onset.

Disturbance observer was adopted in other teleoperation and haptics works as well (e.g., [8, 9, 14, 20]), however, mostly to compensate for inter-axis inertial dynamic coupling or unwanted external forces (e.g., friction) rather than to estimate human forcing. In [9], the same type of disturbance observer [3] as used in this paper was employed for human force estimation, however, the results of [9] is formulated only for one-dimensional point-mass type virtual

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proxy, thus, it is unclear how their results can be extended/applied to multi-DOF nonlinear virtual proxy as considered in this paper.

The rest of this paper is organized as follows. Virtual coupling technique for multi-DOF nonlinear virtual proxy, as presented in [17], is reviewed in Sec. 2 along with its limitations for the contact tasks. Passive decomposition is also briefly summarized in Sec. 2. A novel transparent virtual coupling, which incorporates feedforward action designed through the passive decomposition, is then presented in Sec. 3, along with a brief study of parametric uncertainty effect. The human force estimation algorithm and its modification to combat with the presence of device control-torque saturation is also addressed in Sec. 3. A preliminary experimental investigation of the proposed framework is then performed in Sec. 4, followed by some concluding remarks in Sec. 5.

2 PRELIMINARY

2.1 Virtual Coupling and its Limitations

A typical virtual coupling consists solely of kinematic feedback actions, namely virtual spring and virtual damper. Residing between the discrete-time virtual proxy (VP) and the continuous-time haptic device (HD), this virtual coupling is a hybrid-time system. As an example, here, we give the expression of the virtual coupling presented in [17] for the multi-DOF nonlinear Lagrangian HD and VP, assuming that the master HD and the slave VP are kinematically similar: for $t \in [t_k, t_{k+1})$ with $T_k = t_{k+1} - t_k > 0$,

$$\tau(t) := -b_d\dot{y}(t) - B_{vc}(\frac{y_k - y_{k-1}}{T_{k-1}} - \hat{v}_{k-1}) - K_{vc}(y_k - x_k)$$
$$\tau_k := -b_v\hat{v}_k - B_{vc}(\hat{v}_k - \frac{y_k - y_{k-1}}{T_{k-1}}) - K_{vc}(\hat{x}_k - y_k) \quad (1)$$

where $\tau(t), \tau_k \in R^n$ are the virtual coupling controls respectively for the HD and the VP; $x_k, v_k \in R^n$ are the VP's configuration and velocity, and $y(t) \in R^n$ is the HD's configuration with $y_k \in R^n$ being its sampled at t_k; $B_{vc}, K_{vc} \in R^{n\times n}$ are the damping and spring gains of the virtual coupling; and $b_d \in R^{n\times n}$ is the (minimum) device physical damping and $b_v \in R^{n\times n}$ serves a similar dissipating role in virtual world as b_d.

In [17], the VP is implemented by using non-iterative passive mechanical integrator (NPMI [16]), which then enforces discrete-time passivity of the VP (for a brief description, see Sec. 3). This passive simulation allows for a complete device-simulation separation [2], that is, we can choose the VP parameters completely independently from the device servo-loop (e.g., sampling rate, device damping, etc.). This NPMI is implicit, yet, non-iterative, thus, can still be solved haptically fast while guaranteeing discrete-time passivity. Due to this NPMI integration, the discrete-time virtual damper b_v in (1) functions as a dissipating energy in the discrete-time domain, being coupled to $\hat{v}_k := \frac{v_{k+1}+v_k}{2}$, which is the "representative" velocity for the NPMI implementation. Similar "representative" configuration from the NPMI, $\hat{x}_k = \frac{x_{k+1}+x_k}{2}$ is also used in (1) to enforce passivity. For more details on this NPMI, refer to [16, 17].

Even though, in [17], the NPMI ensures discrete-time passivity of the nonlinear Lagrangian VP, since the virtual coupling (1) is a hybrid-time system with some terms (e.g., $\hat{v}_{k-1}$, $y_k - y_{k-1}/T_{k-1}$) being delayed, we have the following Colgate-like conditions for the passivity of the virtual coupling (1), as shown in [17]:

$$b_d \geq B_{vc}(1 + \frac{T_k}{T_{k-1}}) + K_{vc}T_k$$
$$b_v \geq \frac{B_{vc}}{2}(\frac{T_k}{T_{k-1}} - 1) + \frac{K_{vc}T_k}{2} \quad (2)$$

implying that we cannot increase the virtual coupling gains B_{vc}, K_{vc} indefinitely to improve its transparency and their maximums are rather bounded due to the device damping b_d and the VP local absolute damping b_v. With this condition (2), a passive master and slave coordination at $x_k - y_k \to 0$ is achieved unless there exist any external forces.

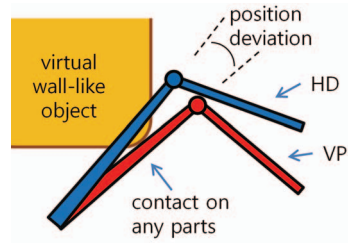

Figure 2: Virtual coupling necessitates position-deviation to produce haptic feedback.

Now, suppose that the VP is maintaining a contact against a virtual object (e.g. virtual wall) with a certain amount of contact force. In this case, the user would like to feel the contact force same as that between the VP and the virtual object. However, the haptic feedback to the user is generated only by the deformation in the spring and damping connection between the HD and the VP. So, if the velocities of the HD and the VP are small, as usually true for contact tasks, the haptic feedback will then be mainly generated by the spring deformation. This means the HD-VP position coordination error, which degrades transparency, should always happen to provide haptic feedback whenever a contact exists. See Fig. 2.

This HD-VP position deviation in virtual coupling comes from its solely relying on the kinematic feedback action to generate haptic feedback. Although we may want to choose the larger K_{vc} of the virtual coupling to improve the transparency, the K_{vc} typically has a limitation because of the passivity conditions like (2). So, alternatively, we utilize passive decomposition [18, 19] to design a suitable feedforward action while adequately taking into account the nonlinear dynamics of the multi-DOF VP and HD as explained in the next Sec. 2.2.

2.2 The Passive Decomposition [18, 19]

The main idea of passive decomposition is to decompose the total $2n$-DOF nonlinear Lagrangian dynamics into two n-DOF *shape system* and *locked system* while preserving open-loop passivity. In the context of master-slave systems, the *shape system* represents the master-slave coordination aspect, while the *locked system* the overall motion of the coordinated master and slave systems. This passive decomposition was successfully applied to nonlinear teleoperation in [18]. Here, we will briefly review this passive decomposition and its application to the teleoperation as presented in [18]. In Sec. 3, we will then extend this result to the hybrid-time haptic system, consisting of multi-DOF (possibly nonlinear) continuous-time haptic device and discrete-time virtual proxy.

Let us consider the following teleoperator system, consisting of two n-DOF nonlinear Lagrangian master and slave robots:

$$M_1(q_1)\ddot{q}_1 + C_1(q_1, \dot{q}_1)\dot{q}_1 = T_1 + F_1 \quad (3)$$
$$M_2(q_2)\ddot{q}_2 + C_2(q_2, \dot{q}_2)\dot{q}_2 = T_2 + F_2 \quad (4)$$

where $q_i, T_i, F_i \in R^n$ are the configuration, the control, and the user/environmental interaction force; $M_i(q_i) \in R^{n\times n}$ are the symmetric and positive definite inertia matrices; $C_i(q_i, \dot{q}_i) \in R^{n\times n}$ are the Coriolis matrices, for the master ($i = 1$) and the slave ($i = 2$), respectively. It is then well known that $\dot{M}_i(q_i) - 2C_i(q_i, \dot{q}_i)$ is skew symmetric, or, equivalently, the master and the slave dynamics are individually passive s.t., $\forall T \geq 0$,

$$\int_0^T [F_i + T_i]^T \dot{q}_i dt = \kappa_i(T) - \kappa_i(0) \geq -\kappa_i(0)$$

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where $\kappa_i(t) := \frac{1}{2}\dot{q}_i^T M_i(q_i)\dot{q}_i$ is the storage function.

Following [18], passive decomposition for (3)-(4) is then given by: with $q_E = q_1 - q_2$,

$$\begin{pmatrix} v_L \\ v_E \end{pmatrix} = S(q)\begin{pmatrix} \dot{q}_1 \\ \dot{q}_2 \end{pmatrix} = \begin{bmatrix} I-\phi(q) & \phi(q) \\ I & -I \end{bmatrix}\begin{pmatrix} \dot{q}_1 \\ \dot{q}_2 \end{pmatrix} \tag{5}$$

where $v_L, v_E := \dot{q}_E \in \Re^n$ are respectively the locked and shape system velocities, $\phi(q) := [M_1(q_1)+M_2(q_2)]^{-1}M_2(q_2)$, and $S(q) \in \Re^{2n\times 2n}$ is the non-singular decomposition matrix. The compatible transform of (5) is also defined to by

$$\begin{pmatrix} T_L \\ T_E \end{pmatrix} = S^{-T}\begin{pmatrix} T_1 \\ T_2 \end{pmatrix}, \quad \begin{pmatrix} F_L \\ F_E \end{pmatrix} = S^{-T}\begin{pmatrix} F_1 \\ F_2 \end{pmatrix} \tag{6}$$

where T_L, F_L and T_E, F_E are the transformed control torque and external force for the locked and shape systems.

Then, using (5)-(6), we can rewrite the $2n$-DOF master-slave dynamics (3)-(4) s.t.

$$M_L\dot{v}_L + C_L v_L + C_{LE}v_E = T_L + F_L \tag{7}$$

$$M_E\dot{v}_E + C_E v_E + C_{EL}v_L = T_E + F_E \tag{8}$$

where the terms with M_L, C_L and M_E, C_E defines the locked and shape systems' dynamics, with $M_L(q) := M_1(q_1)+M_2(q_2)$ and $M_E(q) := \phi^T M_1\phi + [\phi^T - I]M_2[\phi - I]$ being their respective positive-definite symmetric inertias and $\dot{M}_L - 2C_L, \dot{M}_E - 2C_E$ being both skew-symmetric (i.e., the locked and shape systems are individually passive); and $C_{LE}v_E, C_{EL}v_L$ are skew-symmetric (i.e., passive locked-shape coupling with $C_{LE} + C_{EL}^T = 0$). For more details, see [18, 19].

In [18], the following controls are designed for the locked and shape systems (7)-(8):

$$T_L := C_{LE}v_E + \frac{1-\eta}{\eta}F_L \tag{9}$$

$$T_E := C_{EL}v_L - K_v v_E - K_p q_E - F_E \tag{10}$$

where the first terms in (9) and (10) are to eliminate the coupling terms in (7)-(8); $K_v, K_p \in R^{n\times n}$ are the symmetric and positive definite damping and spring gains to stabilize, together with the feedforward cancelation of the non-coordinating force F_E, the shape system to attain $q_E = q_1 - q_2 \to 0$; and $\eta > 0$ is to scale the apparent inertia M_L of the coordinated master and slave systems by modulating the locked system to be

$$\eta \cdot [M_L(q)\dot{v}_L + C_L(q,\dot{q})v_L)] = F_L \tag{11}$$

with $\eta < 1$ implying the scaling-down the apparent inertia. In the next Sec. 3, we will extend this passive decomposition based controls (9)-(10) to the continuous-time haptic device and the discrete-time virtual proxy to improve the transparency of the virtual coupling.

3 TRANSPARENT VIRTUAL COUPLING VIA PASSIVE DECOMPOSITION

Different from Sec. 2.2, the haptic system is a hybrid-time system, consisting of a continuous-time haptic device (HD) and a discrete-time virtual proxy (VP). We model the HD using the n-DOF nonlinear Lagrangian dynamics (3) and simulate a n-DOF nonlinear Lagrangian VP (see Fig. 1) by utilizing NPMI. More precisely, the NPMI expression for the n-DOF VP to emulate (4) is given by [17] : during $T_k := [t_k, t_{k+1})$,

$$I\frac{v_{2,k+1}-v_{2,k}}{T_k} + Q_{2,k}\frac{v_{2,k+1}+v_{2,k}}{2} = u_k + \delta_k \tag{12}$$

$$\frac{q_{2,k+1}-q_{2,k}}{T_k} = M_{2,k}^{-\frac{1}{2}}\frac{v_{2,k+1}+v_{2,k}}{2} \tag{13}$$

where $q_{2,k} \in R^n$ is the VP's configuration with $M_{2,k} = M_2(q_2(t_k))$; $v_k \in \Re^n$ is the "transformed" velocity related to $q_{2,k}$ through the kinematic relation (13); $u_k := M_{2,k}^{-1/2}T_{2,k}, \delta_k := M_{2,k}^{-1/2}F_{2,k} \in R^n$ are the "transformed" control and external force with $T_{2,k}, F_{2,k}$ being the original control and force in (4); and $Q_{2,k} \in R^{n\times n}$ is transformed Coriolis matrix as given by

$$Q_{2,k} := M_{2,k}^{-\frac{1}{2}}C_{2,k}M_{2,k}^{-\frac{1}{2}} - \dot{M}_{2,k}^{\frac{1}{2}}M_{2,k}^{-\frac{1}{2}} \tag{14}$$

which is skew-symmetric with $C_{2,k} := C_2(q_2(t_k), \dot{q}_2(t_k))$. We can then show that the NPMI implementation (12)-(13) possesses discrete-time passivity s.t.:

$$\sum_{k=1}^{N}\hat{v}_{2,k}^T[u_k + \delta_k]T_k = \kappa_{2,N} - \kappa_{2,0} \tag{15}$$

where $\hat{v}_k := (v_{k+1}+v_k)/2$ and $\kappa_{2,k} := \frac{1}{2}v_{2,k}^T I v_{2,k} = \frac{1}{2}v_{2,k}^T M_{2,k}v_{2,k}$ is the kinetic energy. Thanks to this discrete-time passivity, we can render a "light" virtual proxy. See [16, 17] for more details on the NPMI and its properties.

To attain the transparent coupling between the n-DOF (possibly nonlinear) HD (3) and VP (12), we would then like to emulate the control actions (9)-(10) between them, the latter for enforcing the HD-VP coordination and the former for scaling down (or up) the apparent inertia of the coordinated HD and VP. Recall that this inertia scaling is for the locked system dynamics, that is, not only for VP but also for HD, which implies that we can make a locked system even lighter than HD itself. For this controls, let us first decode (9)-(10) into T_1 and T_2 s.t.

$$\begin{aligned} T_1 = &(\frac{1-\eta}{\eta}I - \frac{1}{\eta}\phi^T)(F_1+F_2) + F_2 \\ &+ (I-\phi^T)C_{LE}v_E + C_{EL}v_L - K_v v_E - K_p q_E \end{aligned} \tag{16}$$

$$\begin{aligned} T_2 = &\frac{1}{\eta}\phi^T(F_1+F_2) - F_2 \\ &+ \phi^T C_{LE}v_E - C_{EL}v_L + K_v v_E + K_p q_E \end{aligned} \tag{17}$$

These decoded controls (16)-(17) are for the continuous-time master and slave systems (3)-(4). Thus, we modify (16)-(17) in such a way that they can be implemented between the continuous-time HD and the discrete-time VP s.t.: for $t \in [t_k, t_{k+1})$ and $T_k = t_{k+1} - t_k$,

$$\begin{aligned} T_1(t) = &-b_d\dot{q}_1(t) + (\frac{1-\eta}{\eta}I - \frac{1}{\eta}\phi_k^T)(F_{1,k}+F_{2,k}) + F_{2,k} \\ &+ ((I-\phi_k^T)C_{LE,k} + C_{EL,k}(I-\phi_k) - B_{vc})\frac{q_{1,k}-q_{1,k-1}}{T_{k-1}} \\ &+ ((I-\phi_k^T)C_{LE,k} + C_{EL,k}\phi_k + B_{vc})\hat{v}_{2,k-1} - K_{vc}(q_{1,k}-q_{2,k}) \end{aligned} \tag{18}$$

$$\begin{aligned} T_{2,k} = &-b_v\hat{v}_{2,k} + \frac{1}{\eta}\phi_k^T(F_{1,k}+F_{2,k}) - F_{2,k} \\ &+ (\phi_k^T C_{LE,k} - C_{EL,k}(I-\phi_k) + B_{vc})\frac{q_{1,k}-q_{1,k-1}}{T_{k-1}} \\ &+ (\phi_k^T C_{LE,k} - C_{EL,k}\phi_k - B_{vc})\hat{v}_{2,k} - K_{vc}(\hat{q}_{2,k} - q_{1,k}) \end{aligned} \tag{19}$$

where $\hat{v}_{2,k} = \frac{q_{2,k+1}-q_{2,k}}{T_k}$ and $\hat{q}_{2,k} = \frac{q_{2,k+1}+q_{2,k}}{2}$ following the NPMI implementation [17]; $\phi_k, C_{LE,k}, C_{EL,k}$ are computed similar to Sec. 2.2 with the terms there replaced by their discrete-time counterparts; and $F_{i,k} := F_i(t_k)$, $i = 1,2$.

In (18)-(19), the terms with b_d, b_v, B_{vc}, K_{vc} are the standard virtual coupling as given in (1), whose (hybrid-time two-port) passivity can then be granted by the conditions (2). On the other hand, the

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rest of (18)-(19) in general does not preserve passivity: we may enforce their passivity by using the fictitious flywheel [18] or PO/PC-type mechanism [10] though, which will be reported in a future publication. Even if the passivity of (18)-(19) is not granted and, consequently, limits some freedom of our choosing the paraments of the VP, we found the effect of those extra "feedforward" control actions, derived from the passive decomposition (16)-(17) for the nonlinear multi-DOF HD and VP, can significantly improve the transparency of the standard virtual coupling (1). See the experimental results in Sec. 4.

On the other hands, to use (18) and (19), we should know ϕ, which contains uncertain inertia parameters of HD. Note that, inertia parameters of VP are always available. To investigate the effect of this uncertainty, recall the equation (3) and (4):

$$\begin{pmatrix} M_1(q_1)\ddot{q}_1 + C_1(q_1,\dot{q}_1)\dot{q}_1 \\ M_2(q_2)\ddot{q}_2 + C_2(q_2,\dot{q}_2)\dot{q}_2 \end{pmatrix} = \hat{S}^T \begin{pmatrix} T_L \\ T_E \end{pmatrix} + \begin{pmatrix} F_1 \\ F_2 \end{pmatrix} \tag{20}$$

where $\hat{S}^T$ is used for estimated parameters. With the real decomposition S^{-T}, (20) changes into:

$$\begin{pmatrix} M_L\dot{v}_L + C_L v_L + C_{LE} v_E \\ M_E\dot{v}_E + C_E v_E + C_{EL} v_L \end{pmatrix} = S^{-T}\hat{S}^T \begin{pmatrix} T_L \\ T_E \end{pmatrix} + \begin{pmatrix} F_L \\ F_E \end{pmatrix} \tag{21}$$

where T_L and T_E are in (9) and (10).

Here, we assume that the C_{LE} and C_{EL} are ignorable. This is usually true because the coupling terms are quadratic in velocity and the velocity is small in contact. With this assumption, shape dynamics (8) changes:

$$M_E\ddot{q}_E + C_E\dot{q}_E = \frac{1}{\eta}(\phi^T - \hat{\phi}^T)F_L - K_v\dot{q}_E - K_p q_E \tag{22}$$

where $\hat{\phi}$ is a estimated value. Now, this (22) is same as (8) if $\hat{\phi}$ equals ϕ, which cannot be happen in general. However, at a static contact, F_L in (11) goes to zero since $\dot{q}_i, \ddot{q}_i$ are both zero. This is because, in (6), F_L is defined as $F_1 + F_2$, which is independant to the inertia parameters of the HD. This then means (22) has the same equilibrium point with (8) at origin during the contact. That is, even though we have an uncertainty in parameters, our proposed control (18)-(19) can improve transparency especially in a contact.

Now, notice that, to implement this new virtual coupling control (18)-(19), we need to measure the human force F_1 and the VP's contact force F_2. The VP force F_2 is always available from the simulation (12)-(13). However, measuring the human force F_1 requires some force sensor, which is typically missing in most of the commercially-available haptic devices (e.g., Force Dimension Omega3®, Geomagic Phantom Omni®, Novint Falcon®). To circumvent this, we utilize nonlinear disturbance observer of [3] to (sensorlessly) estimate human interaction force F_1 in the following section.

3.1 Human Force Observer with Device Saturation

To estimate the human force for the haptic device (3), we utilize the nonlinear disturbance observer proposed in [3]. For this, considering the human force F_1 as the disturbance d (i.e., $d := F_1$), we rewrite (3) s.t.

$$d = M(q)\ddot{q} + C(q,\dot{q})\dot{q} - T \tag{23}$$

where the subscripts are omitted for brevity. The estimate of d is then given by [3]

$$\hat{d}(t) := z(t) + p(q,\dot{q}) \tag{24}$$

where $z \in R^n$ is an auxiliary variable, which is updated according to

$$\dot{z} = -L(q,\dot{q})z + L(q,\dot{q})\left[C(q,\dot{q})\dot{q} - T - p(q,\dot{q})\right] \tag{25}$$

where the control input T computed from (17), which may be different from the real control T(t). Here $L(q,\dot{q}) \in R^{n\times n}$ and $p \in R^n$ are chosen s.t.

$$L(q,\dot{q})M(q)\ddot{q} = \begin{bmatrix} \frac{\partial p(q,\dot{q})}{\partial q} & \frac{\partial p(q,\dot{q})}{\partial \dot{q}} \end{bmatrix} \begin{bmatrix} \dot{q} \\ \ddot{q} \end{bmatrix} = \dot{p}$$

then, the error dynamics becomes

$$\dot{e} := -L(q,\dot{q})e \tag{26}$$

where $e := d - \hat{d}$, with a assumption that d to be constant (or slowly-varying). Thus, if we choose L in (26) as asymptotically stable, $\hat{d}$ converges to d [3].

For the two-link revolute manipulator as used for our experiments in Sec. 4 (i.e. $q = [q_1, q_2]^T$), its inertia matrix is:

$$M(q) = \begin{bmatrix} Y_1 + 2X\cos q_2 & Y_2 + X\cos q_2 \\ Y_2 + X\cos q_2 & Y_3 \end{bmatrix}$$

then, p and L can be computed by

$$p(q,\dot{q}) = c\begin{pmatrix} \dot{q}_1 \\ \dot{q}_1 + \dot{q}_2 \end{pmatrix}, \quad L(q,\dot{q}) = c\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} M^{-1}(q)$$

where c is chosen s.t. $c \geq X\dot{q}_{2m}$ [3].

Finding p and L is an important part to complete observer design in (25). Though it is not easy, a simple design way is presented in [9]. Besides, a design method which is based on linear matrix inequality are developed in [21]. With this method, we can find a proper p fairly easy even though the device dynamics is complicated (i.e., more than 2-DOF haptic device). See [21] for more details.

Another problem we encountered with the force observer (25) is the "wind-up" problem when the haptic device control torque is saturated. To better see this problem, consider a static contact. To make the exposition simpler, consider also only one axis of the haptic device (3). Denote the maximum control torque by T_m and the commanded control input by T_c. Then, the real device control torque is given by

$$T = \begin{cases} T_m & if\ \ T_c \geq T_m \\ T_c & if\ \ T_c \leq T_m \end{cases} \tag{27}$$

Now, suppose that the device torque is saturated, yet, this saturation is not reflected in the force estimation (25). This then implies that, instead of the real $T = T_m$, the observer instead uses $T = T_c$ for the estimation (25). That is, from (23), the real disturbance d is given by $d = -T_m$. However, from (24)-(25), $\hat{d}$ is given as $\hat{d} = -T_c$ with $\dot{z} = 0$. So, combining these two equations, we can then obtain

$$\hat{d} = F_1 + (T_m - T_c)$$

with $d = F_1$. This then shows that the observer 'feels' the device saturation effect as 'disturbance'. Consequently, the d from (25) is overestimated by the magnitude of $T_m - T_c$. Here, note that, in static contact, the signs of F_1 and T are opposite (e.g., F_1 and $T_m - T_c$ both negative).

Therefore, in the presence of device control-torque saturation, the force estimation (25) will over-estimate the human force, essentially increasing the feedforward terms in T_1 and T_2 (18)-(19). Since those feedforward terms are for perfect coordination and the device is already saturated, the HD stays where it is, while the VP starts moving toward the HD position. This then means that, during this static contact, the VP will lurk into, e.g., the virtual wall of Fig. 2, which is not realistic. During this process, the estimated force $\hat{d}$

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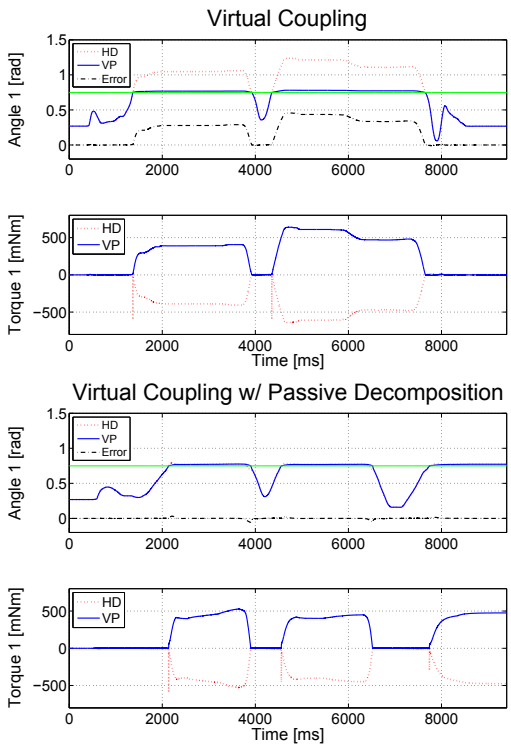

Figure 3: Contact task with standard virtual coupling and our proposed framework

may also be accumulated to be a large value. Then, when the human user tries to move the HD out from the virtual wall, $\hat{d}$ would still be large, thereby, generating an abnormally large force pushing the haptic device even faster from the virtual wall.

To prevent this "wind-up" phenomena, we incorporate the device saturation information in the force estimator (25). That is, since the reason of this is because of using T_c rather than T_m, we inject T_m to (25) whenever $T_c > T_m$. This strategy we found is fairly effective to reduce the wind-up problem of the force estimator in the presence of the device control-torque saturation. See Sec. 4. This saturation problem was also considered in [9] by projecting the output $\hat{d}$ directly below T_m.

4 PRELIMINARY EXPERIMENTAL RESULTS

We adopted Phantom Omni® as the haptic device. We constrained one of its actuated-DOF and utilized only the remaining two-DOF. A 2-DOF nonlinear Lagrangian virtual proxy is also implemented with the following parameters: $(m_{v1}, m_{v2}) = (0.00167, 0.001)$[kg], $(I_{v1}, I_{v2}) = (2.5589 \times 10^{-4}, 6.4094 \times 10^{-5})$[kgm^2], and $(l_{v1}, l_{v2}) = (1.356, 0.877)$[m], where $m_\star, I_\star, l_\star$ are the mass, the moment of inertia, and the length of the proximal and distal links. The least-square estimation is used for the haptic device inertia parameters s.t.: $(m_{h1}, m_{h2}) = (0.05352, 0.03277)$[kg], $(I_{h1}, I_{h2}) = (8.2 \times 10^{-5}, 2.1 \times 10^{-5})$[kgm^2], and $(l_{h1}, l_{h2}) = (0.1356, 0.0877)$[m]. In addition, we set the gains for the controls (18)-(19) s.t.: $B_{vc} = \mathrm{diag}[0.001, 0.0008]$[Nms], $K_{vc} = \mathrm{diag}[0.14, 0.12]$[Nm], $b_v = \mathrm{diag}[0.0015, 0.0013]$[Nms]. Also, we rendered a wall-like virtual object which can contact on first link at $q_1 > 0.75$[rad], with $B_o = 0.03$[Nms] and $K_o = 20$[Nm] for contact tasks. See Fig. 2. The disturbance observer of Sec. 3.1 was also used with $c = 0.035$ which satisfies the condition stated in Sec. 3.1. For the controls (18)-(19), we also omitted the terms with $C_{LE,k}$ and $C_{EL,k}$ since: 1) thanks to the NPMI implementation, the inertia of the VP can be rendered small enough, thus, these coupling effects, which stem

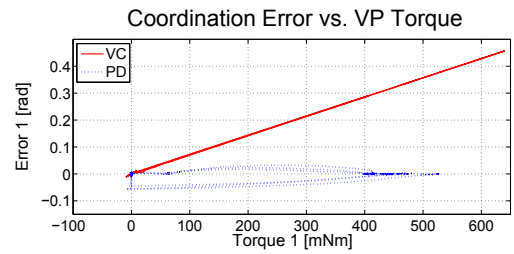

Figure 4: Coordination performance of standard virtual coupling and our proposed framework

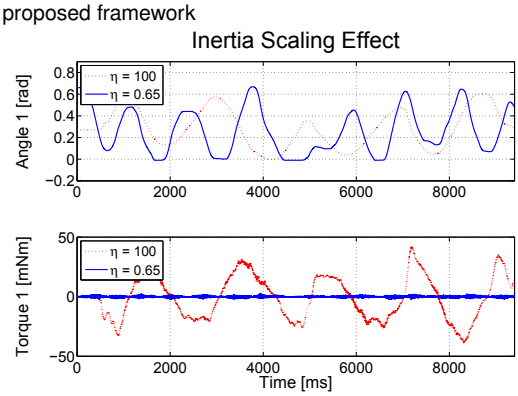

Figure 5: Effect of different inertia scaling factor η

from the inertial coupling, are already small; and 2) during the contact tasks (with small velocity), the coupling terms in (7)-(8), which are quadratic in velocity, also become small.

The experimental results, with the standard virtual coupling and our passive decomposition based framework, are presented in Fig. 3. For contact tasks, it is clear that, due to the feedforward action, our proposed framework can maintain much more precise position coordination between nonlinear multi-DOF haptic device and virtual proxy, thereby, substantially improves transparency of the haptic interaction. Notice also that we use impulsive forces at the beginning of the contacts to enhance the contact perception [7]. In Fig. 4, we also plot the position coordination error versus haptic device control force during several contact tasks (including approaching/detaching process as well). From there, we can even clearly see the efficacy of our proposed passive decomposition based framework: for the standard virtual coupling, the coordination error increases proportionally to the control force. However, for our proposed framework, the coordination error is maintained small *regardless* of the operating force. The small deviation in Fig. 4 mainly results from observer estimation time. For example, at the very point to moving outside, T_1 is zero with some position deviation because the control from spring force is cancelled by the remaining estimated force. However, at a quasi-static contact in large force (e.g. $T_1 > 400$ in Fig. 4), the error converges to zero.

Although, thanks to NPMI implementation (12), we can render a "light" VP, for some applications, we may want the coordinated haptic device and virtual proxy lighter than haptic device. For this, we may utilize the inertia scaling η in (18)-(19) for virtually scaling down the apparent inertia of the combined VP and HD. Fig. 5 contains the results with this inertia scaling during "shaking" experiment. From there, we can see that, with $\eta < 1$ (i.e., inertia scaled-down), the human user could perceive the combined VP and HD as light (i.e., very small human forcing), whereas with $\eta > 1$ (i.e., inertia scaled-up, used for, e.g., rehabilitation), as heavy.

We also performed experiments with and without the projection strategy of Sec. 3.1 to combat with the device control-torque saturation and the results are given in Figs. 6 and 7. From there, we can

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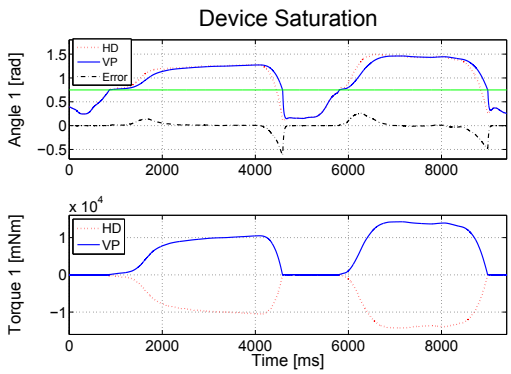

Figure 6: Device saturation effect without projection strategy

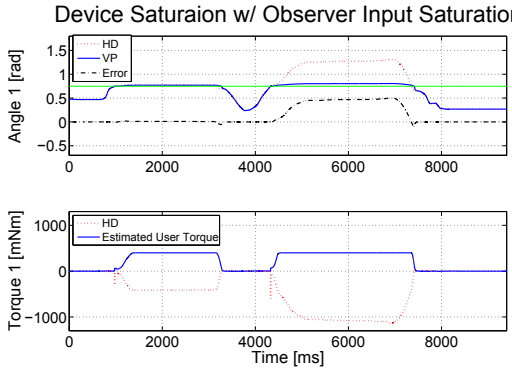

Figure 7: Device saturation effect with projection strategy

then see that: 1) without the strategy, the virtual proxy lurked into the virtual object while the estimated force was accumulated, and, when the device released, it was driven by the accumulated force further out from the object (see the sudden increase of the position error in Fig. 6), however, 2) with the strategy, such a "wind-up" problem disappeared and the human could still sharply perceive the contact with the virtual object, although there is some position deviation between HD and VP. Note that, if there were no device saturation, such a position deviation would be gone with sharp force reflection as in Fig. 3.

5 CONCLUSION

In this paper, we present a novel transparent virtual coupling framework between multi-DOF and continuous-time haptic device and discrete-time virtual proxy. Our proposed framework significantly enhances the transparency of the standard (purely feedback-based) virtual coupling technique, particularly during contact tasks, by utilizing some feedforward actions, which allow for a precise device-proxy coordination while also providing an ability to scaling down/up the apparent inertia of the coordinated device and proxy. Passive decomposition is used and extended to address hybrid-time control between the haptic device and the virtual proxy. Nonlinear disturbance observer is also adopted to estimate human force, along with a certain projection strategy to combat with the problem of device saturation. Some preliminary experimental results are also presented to illustrate the proposed framework.

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